



COURSE HANDOUT (PART II)
First Semester 2026-27

Course No.	CS F372
Course Title	Operating Systems
Instructor-in-Charge	Prof. Avinash Gautam (avinash@pilani.bits-pilani.ac.in)
Instructor	Prof. J. P. Misra (jpm@@pilani.bits-pilani.ac.in)
Tutorial Instructors	Prof. Yashvardhan Sharma (yash@@pilani.bits-pilani.ac.in) Ms. Neha (p20240030@pilani.bits-pilani.ac.in) Mr. M VISHNU CHAITANYA (p20240488@pilani.bits-pilani.ac.in)
Handout Date	01 August 2026

In addition to Part I (General Handout for all courses appended to the timetable), this part provides further course-specific details.

1. Scope and Objectives of the Course:

- To provide an understanding of the functions of operating systems.
- To provide insight into the functional modules of operating systems.
- To study the concepts underlying the design and implementation of operating systems.

2. Text Book:

T1. A. Silberschatz and P. B. Galvin, *Operating System Concepts*, 7th/8th Edition, Addison-Wesley, 2005.

3. Reference Book:

R1. W. Stallings, *Operating Systems*, 4th Edition, PHI, 2001.

4. Course Plan

Module Number	Lecture Session	Reference	Learning Outcome / Topic
1. Introduction and System Structures (06 Lectures)	L1	T1: 1.1-1.3	Introduction to Operating Systems
	L2-L3	T1: 1.4-1.9	OS architecture and its components
	L4	T1: 1.10-1.12	Special-purpose systems
	L5-L6	T1: Chapter 2	System structures
2. Processes and CPU Scheduling (08 Lectures)	L7-L9	T1: Chapter 3	Process concept, scheduling, operations on processes and inter-process communication
	L10-L14	T1: Chapter 5	Process-scheduling algorithms
3. Synchronization and Deadlocks (11 Lectures)	L15-L16	T1: 6.1-6.3	Process synchronization and the critical-section problem
	L17-L19	T1: 6.4-6.9	Semaphores, monitors and synchronization hardware
	L20-L21	T1: 7.1-7.3	Deadlock concepts and characterization
	L22-L24	T1: 7.4-7.6	Deadlock detection, prevention and avoidance
	L25	T1: 7.7	Recovery from deadlocks
4. Memory Management (12 Lectures)	L26-L27	T1: 8.1-8.3	Memory management and allocation
	L28-L30	T1: 8.6; class notes	Segmentation and hardware support in Intel processors
	L31-L33	T1: 8.4-8.5	Paging and hardware support in Intel processors
	L34	T1: 9.1-9.3	Virtual-memory management
	L35-L36	T1: 9.4	Page-replacement algorithms
	L37	T1: 9.5-9.6	Frame allocation and thrashing



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Module Number	Lecture Session	Reference	Learning Outcome / Topic
5. Storage and File Systems (05 Lectures)	L38-L39	T1: Chapter 12	Secondary-storage structure and management
	L40-L42	T1: Chapter 10	File systems

5. Evaluation Scheme:

Component	Duration	Weightage	Date & Time	Remarks
Mid-Semester Test	90 minutes	35%	06/10/2026 (FN1)	Open book
* Tutorial Test – 1	60 minutes	12%	20/09/2026	Closed book
* Tutorial Test – 2	60 minutes	13%	15/11/2026	Closed book
Comprehensive Examination	180 minutes	40%	05/12/2026 (FN)	Closed book

* The timings and other details related to the tutorial tests will be announced soon on Nalanda.

6. Make-Up Policy:

- Makeup will be given only to for the genuine reasons with proper evidence.
- Medicine prescription alone is not a sufficient evidence. The doctor must recommend bed rest.
- No makeup requests will be entertained if the tutorial attendance is below 80%.
- The makeup requests should reach before the exam.
- No makeup requests will be entertained after the evaluation component is over.

7. Chamber Consultation Hour: Send email to take appointment

8. Notice: All notices concerning this course will be displayed on Nalanda.

9. NC Policy: A student scoring less than 10% marks will be awarded NC.

Instructor-in-Charge
CS F372



FIRST SEMESTER 2026-2027

Tutorial Handout

DATE: 03-08-2026

In addition to course handout this portion gives specific details regarding what will be covered in the operating system tutorials:

COURSE NO : CS F372
COURSE TITLE : OPERATING SYSTEMS TUTORIALS

Instructors : T1: Prof. Avinash Gautam [avinash@pilani.bits-pilani.ac.in]
T2, T5: Mr. MALLIPEDDI VISHNU [p20240488@pilani.bits-pilani.ac.in]
T3: Prof. Yashvardhan Sharma [yash@pilani.bits-pilani.ac.in]
T4: Ms. Neha [p20240030@pilani.bits-pilani.ac.in]

1. OBJECTIVE

In addition to regular operating system lectures, in the tutorials you will learn several UNIX system calls related to file management, process management, threads, inter-process communication using signals etc.

2. PRE-REQUISITES

It is assumed that the students have sufficient programming experience in C and good knowledge of pointers.

4. BOOKS

1. W. Richard Stevens & Stephen A. Rago, "Advance Programming in the UNIX Environment", Second Edition, Pearson India.
2. Kay A. Robbins, Steven Robbins, "Unix Systems Programming Communication, Concurrency, and Threads", First Edition, Pearson India.

5. TUTORIAL PLAN

S. No.	Description
1.	System Calls Related to File Manipulation
2.	Process Creation and Execution – Part I
3.	Process Creation and Execution – Part II
4.	Inter-Process Communication (IPC) using Signals
5.	IPC using Pipes
6.	IPC using Shared Memory
7.	Threads Creation and Execution



8.	Threads Synchronization (Mutex & Condition Variables)
9.	Semaphores

5. TUTORIAL PLAN

Week	Day	Date	Day	Date
1	Mon	10-Aug	Wed	12-Aug
2	Mon	17-Aug	Wed	19-Aug
3	Mon	24-Aug	Wed	26-Aug
4	Mon	07-Sep	Wed	09-Sep
5	Mon	14-Sep	Wed	16-Sep
6	Mon	21-Sep	Wed	23-Sep
7	Mon	28-Sep	Wed	30-Sep
8	Mon	12-Oct	Wed	14-Oct
9	Mon	19-Oct	Wed	21-Oct
10	Mon	26-Oct	Wed	28-Oct
11	Mon	02-Nov	Wed	04-Nov

6. EVALUATION SCHEME:

S. No.	Component & Nature	Duration	Weight	Date and Time	Remarks
1	Tutorial Test-1	60 Min	12%	20/09/2026	Closed Book
2	Tutorial Test-2	60 Min	13%	15/11/2026	Closed Book

7. Chamber Consultation:

Mail the concerned tutorial instructors and take appointment

8. Notices:

All notices will be posted on Nalanda by the respective instructors.

9. Make-Up Policy:

Refer to main course handout.

Instructor-in-charge
CS F372 (Operating Systems)